Project Design Phase

Introduction

The project design phase is one of the most important stages in software development. In this phase, the overall structure of the application is planned before the actual coding begins. The design phase helps developers understand how the system will work, how different components will interact, and how the user will interact with the application.

For the Plant Care application, the design phase focused on creating a simple and organized structure that allows users to manage their plants easily. The main goal was to design a system that is easy to use, visually clear, and efficient in helping users take care of their plants.

System Architecture Design

System architecture refers to the overall structure of the application and how different parts of the system are connected. The Plant Care app follows a basic mobile application architecture where the user interacts with the mobile interface and the application processes the data using a database.

The system mainly consists of three main components:

1. User Interface (UI)
The user interface is the part of the application that users see and interact with. It includes screens such as login page, dashboard, plant details page, and reminders section. The design of the interface should be simple so that users can easily navigate through the application.

2. Application Logic
This part handles the main functionality of the app. It manages tasks such as adding plants, setting reminders, retrieving plant information, and processing user inputs.

3. Database System
The database stores important information such as user accounts, plant details, and care schedules. This ensures that the data remains saved even when the user closes the application.

User Interface Design

User interface design focuses on how the application looks and how users interact with it. For the Plant Care application, the UI design was kept simple and clear so that even beginners can use the app without confusion.

Some of the main screens included in the design are:

Login and Registration Screen
This screen allows new users to create an account and existing users to log into the application.

Home Dashboard
The dashboard is the main screen where users can see a list of all their plants along with upcoming care reminders.

Add Plant Screen
This screen allows users to add a new plant by entering details such as plant name, type, and planting date.

Plant Details Screen
This screen shows detailed information about a specific plant including care instructions and watering schedule.

Reminder Screen
This section displays notifications and reminders related to plant care activities.

The interface design focuses on simple buttons, clear text, and easy navigation.

Database Design

Database design is the process of organizing data in a structured way so that it can be stored and accessed efficiently. In the Plant Care application, the database stores information related to users and plants.

Some important data stored in the database include:

- User account information
- Plant names and types
- Plant care schedules
- Reminder notifications
- Plant images

Each user can store multiple plants in their account, and each plant has its own care information and reminders.

System Flow Design

System flow design explains how the user interacts with the application and how the system responds to those actions.

The basic flow of the Plant Care application is as follows:

1. The user opens the application.
2. The user logs in or creates a new account.
3. The user adds plants to their plant list.
4. The application stores plant details in the database.
5. The system generates reminders based on plant care schedules.
6. The user receives notifications for watering or other tasks.

This flow ensures that the user can easily manage plant care activities through the application.

Technology Design

The technologies used in the project are selected based on ease of development and compatibility with mobile devices.

Some technologies planned for the Plant Care application include:

- **Android Studio** for app development
- **Java or Kotlin** programming language
- **Firebase or a database system** for storing user data
- **Mobile notification system** for reminders

These technologies help in creating a reliable and efficient mobile application.

Summary

The design phase helped in planning the structure and functionality of the Plant Care application before development started. By designing the system architecture, user interface, database structure, and application flow, the development process becomes more organized and efficient.

A well-planned design ensures that the final application is easy to use, reliable, and capable of meeting the needs of users who want help managing their plants.